Real-Time Tactile State Prediction via Physics-Informed Cross-Modal Force-Vision Attention

Anonymous Authors

***Abstract*—Optical tactile sensors such as GelSight capture rich contact geometry at 18 Hz, while co-located force sensors (e.g., XELA) provide 48-dimensional force readings at 100 Hz. This 5.5× frequency gap leaves reactive controllers blind to contact events that unfold within a single camera interval. We address *forward tactile image prediction*: given the current GelSight frame g_t and XELA force readings f_{t-1}, f_t, predict the next frame $\hat{g}_{t+1}$ at sub-10 ms latency. We introduce GelXelaCrossAttn, a lightweight cross-attention network in which image patches query force tokens derived from XELA taxel readings, augmented with a physics-informed spatial warping module that decomposes the prediction into an explicit dense displacement field (modeling gel deformation) and a gated illumination residual (correcting lighting and sensor artifacts). On a dataset of 5,489 quasi-static cable-manipulation samples spanning seven cable types, GelXelaCrossAttn achieves 36.67 dB PSNR and 0.941 SSIM at 2.98 ms model-inference latency—a +0.91 dB improvement over the base cross-attention model without physics priors (35.76 dB). The physics-informed variant beats the copy-previous baseline on all six functional metrics (contact center error, dense flow error, contact IoU at two scales, marker position error, marker-region SSIM), demonstrating control-relevant improvements beyond pixel-level quality. Ablation studies show that cross-attention (+1.84 dB) and temporal dynamics (+1.46 dB) are the critical components, while the physics decomposition contributes +1.03 dB over direct delta prediction. We release the dataset, model code, and evaluation suite.**

***Index Terms*—Tactile Sensing, Cross-Modal Prediction, Force-Vision Fusion, Real-Time Inference, Cable Manipulation.**

I. Introduction

TACTILE sensing is central to dexterous manipulation: optical tactile sensors such as GelSight [1] convert surface deformation into dense, high-resolution images of contact geometry, enabling tasks from grasp monitoring to cable following [2]. Yet GelSight cameras run at ∼18 Hz, while compliant contact events in cable manipulation—incipient slip, cable shift, edge loading—can develop over intervals as short as 10–25 ms, i.e., within a single camera frame. Co-located multi-axis force sensors, such as XELA [3], operate at 100 Hz and readily capture these transient dynamics, but their 4×4 spatial resolution is too coarse for pose estimation or contact segmentation.

a) The gap in prior work.: Existing tactile image synthesis methods are passive *interpolators*: they reconstruct intermediate frames from two known boundary frames, conditioning on the start and end state [2], [4]. This requires the future frame, so it cannot be used in a causal real-time control loop. Action-conditioned prediction methods for dexterous manipulation (e.g., ViTacFormer [5]) operate on different sensor modalities and do not target the sub-10 ms latency regime. To our knowledge, no prior work simultaneously achieves (i) causal, single-step forward prediction from an optical-tactile sensor, (ii) cross-modal conditioning on a heterogeneous force sensor, and (iii) inference latency compatible with a 100 Hz reactive-control budget.

Submitted for review.

b) This paper.: We introduce **GelXelaCrossAttn**, a lightweight cross-attention network for real-time GelSight prediction conditioned on XELA force readings. The model tokenizes GelSight image patches and XELA taxel readings into a shared embedding space, then applies two rounds of cross-attention in which image tokens query force tokens. An explicit *anti-lazy* design—no *learned* encoder-to-decoder skip connection—ensures that contact-change regions must be explained by the force-conditioned delta, rather than routed through a learned bypass of the cross-attention pathway. At 0.83 M parameters and 2.98 ms GPU latency, GelXelaCrossAttn achieves 35.76 dB PSNR on a quasi-static cable-manipulation benchmark, occupying the Pareto-optimal point on the accuracy–latency–parameter frontier.

c) Why a benchmark matters.: Existing tactile prediction work evaluates models in isolation, making it hard to attribute gains to architecture choices versus scale. We compare seven model families—CNN, STN (spatial transformer), UNet, force-only CVAE, and the copy-prev trivial baseline, plus our cross-attention proposal—under a shared latency constraint. This reveals that UNet variants (∼31 M parameters) do *not* outperform GelXelaCrossAttn despite 37× more parameters, and that removing visual context causes a 7.71 dB PSNR collapse. We present these results as an *effectiveness demonstration* on within-distribution quasi-static cable manipulation. An ablation study (Sec. IV-D) isolates the contributions of cross-attention depth, the anti-lazy design, and the force input form; it confirms that the force differential $\Delta\mathbf{f}_t$ is the key signal, while overall PSNR is relatively insensitive to the number of attention blocks. Performance on held-out cable types degrades by 5–9 dB, which is the most important limitation of the current work.

d) Contributions.:

1) **GelXelaCrossAttn**: a causal cross-modal force-vision attention network for sub-10 ms forward tactile image prediction. Among all architectures satisfying the latency constraint, it achieves the highest PSNR (35.76 dB, 2.98 ms) and is Pareto-efficient on the accuracy–latency frontier. GelLite (35.46 dB, 1.10 ms) offers a complementary speed–accuracy trade-off, with marginally better contact-localization on some metrics (Sec. IV).
2) **Systematic benchmark**: we compare seven model families on the same dataset under a shared ≤10 ms latency constraint, providing a rigorous reference for future causal tactile prediction work (Sec. IV).

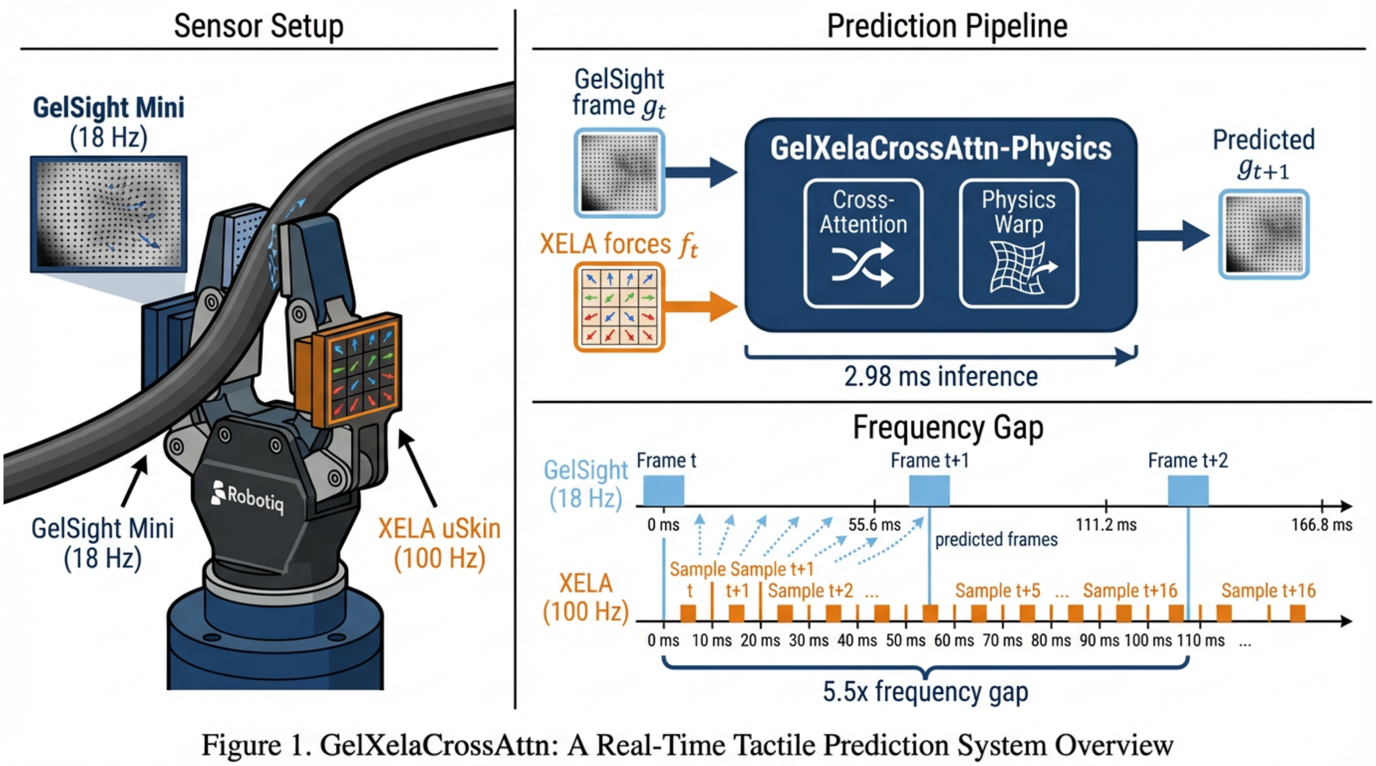

Fig. 1: System overview of GelXelaCrossAttn-Physics. *Left*: Sensor setup—a Robotiq gripper equipped with a GelSight Mini (18 Hz) and an XELA uSkin force array (100 Hz) manipulating a cable. *Center*: The prediction pipeline takes the current GelSight frame g_t and XELA forces $\mathbf{f}_t$ as input, producing the predicted next frame $\hat{g}_{t+1}$ at 2.98 ms inference latency via cross-attention and physics-informed spatial warping. *Bottom-right*: The 5.5× frequency gap between GelSight and XELA, illustrating how predicted frames fill the temporal blind spots.

3) **Dataset and evaluation suite**: 5,489 labeled (prev-frame, force, target-frame) triplets across seven cable types, with pixel-level and contact-level functional metrics, released publicly.

The rest of this paper is organized as follows. Sec. II reviews related work. Sec. III presents GelXelaCrossAttn. Sec. IV reports experiments and analysis. Sec. V concludes.

II. Related Work

a) Optical tactile image synthesis.: A growing body of work generates GelSight-like images from non-visual signals or from paired data across conditions. Wang et al. [6] use GANs to transfer simulated to real tactile images; SimTacLS [7] and related work synthesize deformation patterns from contact geometry. Closest to our setting, TacFlowNet [4] interpolates intermediate GelSight frames from a start frame, end frame, and XELA force readings using bidirectional cross-attention. However, TacFlowNet requires the *future* boundary frame as input, which is unavailable in a causal control loop. Our work addresses this gap: we condition only on past and current observations, enabling real-time forward prediction.

b) Multi-rate sensor fusion.: The asynchronous pairing of a slow, dense sensor with a fast, sparse sensor mirrors the event-camera literature. Time Lens [8] and Time Lens++ [9] use event streams (high temporal resolution, sparse) to interpolate between video frames (low temporal resolution, dense). Our setting is analogous: XELA force readings (100 Hz, 48-dimensional) play the role of event streams, while GelSight frames (18 Hz, 715×551) play the role of video frames. The key difference is that we predict *future* frames rather than intermediate ones, and our signals originate from physically different sensors with no common time-base or pixel-aligned correspondence.

c) Force-conditioned and action-conditioned tactile prediction.: ViTacFormer [5] autoregressively predicts future tactile states from visual-tactile pretraining for dexterous manipulation. Sparsh [10] and ManiFeel [11] learn tactile representations for policy learning, but do not target image-level prediction at inference latency compatible with real-time control. In the cable-manipulation domain, She et al. [2] use GelSight at native 18 Hz with a reactive LQR controller, without any frequency up-sampling or predictive model. Our work is the first to pair causal GelSight prediction from XELA with an explicit sub-10 ms latency constraint, and to provide a systematic multi-architecture benchmark under a shared latency budget.

III. Method

A. Problem Formulation

Let $g_t \in \mathbb{R}^{H\times W}$ be a grayscale GelSight frame at time t (H=715, W=551, normalized to $[-1, 1]$), and let $\mathbf{f}_t \in \mathbb{R}^{48}$ be the co-located XELA force reading (16 taxels × 3 axes:

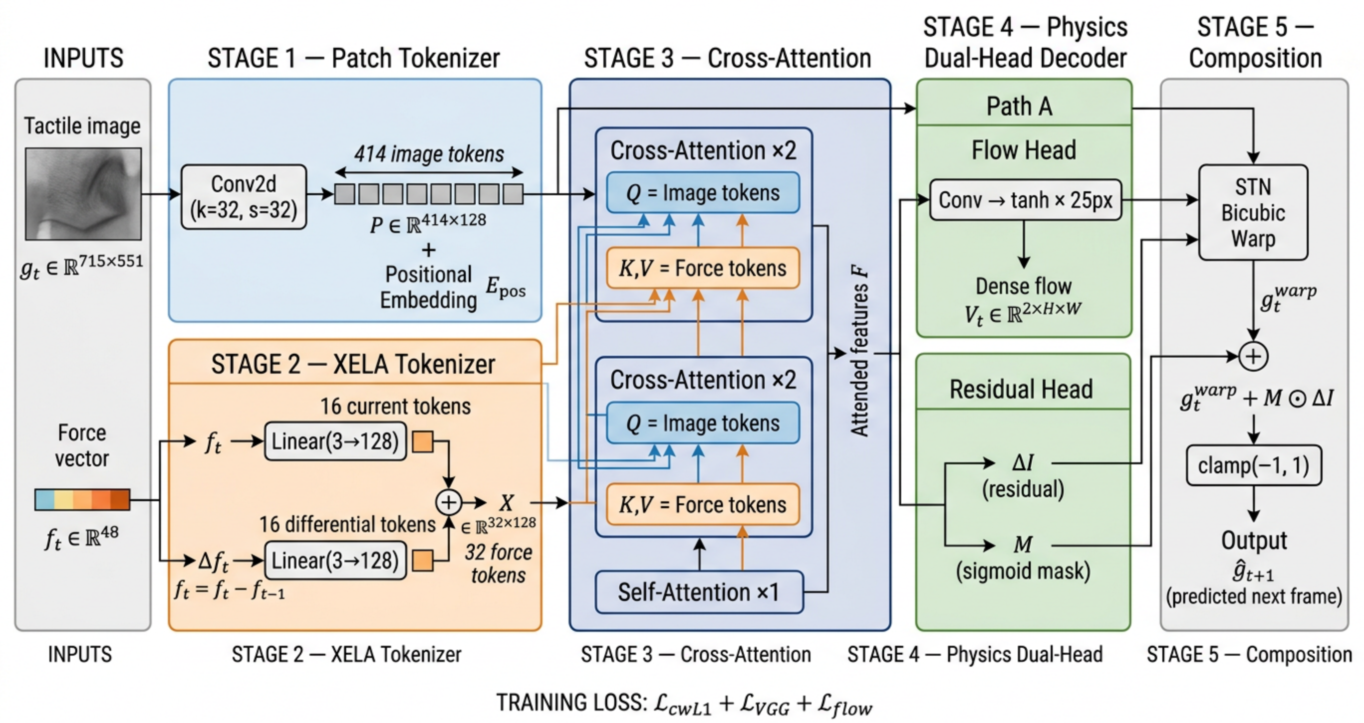

Fig. 2: Architecture of GelXelaCrossAttn-Physics. The GelSight frame g_t is tokenized into 414 image patches via a strided convolution (Stage 1), while XELA force readings are projected into 32 force tokens (Stage 2). Two cross-attention blocks let image tokens query force tokens (Stage 3). A physics-informed dual-head decoder predicts a dense flow field $\mathbf{V}_t$ and a masked illumination residual (Stage 4). The final prediction is composed via bicubic warping and gated residual addition (Stage 5).

F_x, F_y, F_z). Define the force differential $\Delta\mathbf{f}_t = \mathbf{f}_t - \mathbf{f}_{t-1}$. The goal is to learn a *causal* model

$$\hat{g}_{t+1} = \mathcal{M}(g_t,\ \mathbf{f}_t,\ \Delta\mathbf{f}_t) \tag{1}$$

that minimizes $\mathbb{E}[\|g_{t+1} - \hat{g}_{t+1}\|_1]$ subject to an inference latency of ≤ 10 ms per frame (GPU, batch size 1). Using both $\mathbf{f}_t$ (absolute state) and $\Delta\mathbf{f}_t$ (change) gives the model access to the total contact load as well as its time derivative.

B. GelXelaCrossAttn Architecture

The network has five stages described below and illustrated in Fig. 2.

1) Image Patch Tokenizer: A single strided convolution maps the input GelSight frame to a sequence of patch tokens:

$$\mathbf{P} = \mathrm{Conv2d}(g_t;\ 1\to D,\ k{=}32,\ s{=}32) + \mathbf{E}_{\mathrm{pos}} \ \in \mathbb{R}^{N\times D}, \tag{2}$$

where $D{=}128$, $N{=}\lceil H/32\rceil \times \lceil W/32\rceil{=}23{\times}18{=}414$ patches, and $\mathbf{E}_{\mathrm{pos}}$ is a learnable positional embedding. The large stride ($k{=}s{=}32$) keeps N small, which is the primary design choice enabling sub-3 ms latency.

2) XELA Force Tokenizer: Both the force differential $\Delta\mathbf{f}_t$ and the current reading $\mathbf{f}_t$ are reshaped to per-taxel form $[16, 3]$ and projected:

$$\mathbf{X}_\Delta = \mathrm{Linear}(\Delta\mathbf{f}_t \text{ reshaped to } [16,3]) \in \mathbb{R}^{16\times D}, \tag{3}$$

$$\mathbf{X}_c = \mathrm{Linear}(\mathbf{f}_t \text{ reshaped to } [16,3]) \in \mathbb{R}^{16\times D}, \tag{4}$$

with a shared weight Linear$(3\to D)$ applied independently per taxel. Current and differential tokens are concatenated to form 32 force tokens $\mathbf{X} = [\mathbf{X}_\Delta;\ \mathbf{X}_c] \in \mathbb{R}^{32\times D}$, encoding both the absolute contact state and its change.

3) Cross-Attention Blocks: Two cross-attention blocks let image tokens *query* force tokens:

$$\mathbf{P}' = \mathrm{CrossAttn}(\underbrace{\mathbf{P}}_{\text{queries}},\ \underbrace{\mathbf{X}}_{\text{keys, values}}) + \mathbf{P}, \tag{5}$$

with 4 attention heads and residual connections followed by layer normalization. One self-attention block follows each cross-attention block to allow image tokens to propagate information spatially.

a) Anti-lazy design.: Many encoder–decoder architectures include learned skip connections from encoder to decoder. In a tactile prediction network, such skips allow the trivial solution $\hat{g}_{t+1} \approx g_t$ (copy-prev) by routing g_t through the shortcut and ignoring the force tokens entirely. GelXelaCrossAttn has *no learned encoder-to-decoder skip connection*: the feature path from g_t to the decoder runs through the patch tokenizer and the cross-attention blocks, both conditioned on XELA tokens. The final output $\hat{g}_{t+1} = g_t + \delta g$ (Eq. 7) is a *fixed arithmetic residual*, not a learned bypass—equivalent to setting the target as $\delta g = g_{t+1} - g_t$, which is standard in residual image prediction. This distinction matters: without a learned skip, the network cannot learn to trivially copy g_t through a learnable shortcut, so contact-change regions must be accounted for by the force-conditioned δg prediction.

4) Physics-Informed Spatial Warping: Rather than directly predicting an additive delta, we decompose the prediction into a *physics-informed warp* and an *illumination residual*, motivated by the observation that contact-induced gel deformation is dominated by local spatial displacement of the marker pattern.

The attended token map $\mathbf{F}$ is reshaped to a spatial feature map at $\frac{H}{8}\times\frac{W}{8}$ resolution and processed by two parallel heads:

a) (i) Dense flow field.: A convolutional head predicts a displacement field $\mathbf{V}_t \in \mathbb{R}^{2\times H\times W}$ (bilinearly upsampled from $\frac{H}{8}\times\frac{W}{8}$), bounded by $\mathbf{V}_t = d_{\max}\cdot\tanh(\cdot)$ where $d_{\max}$=25 px reflects the physical constraint that quasi-static gel deformation rarely exceeds $\sim$15 px. A spatial transformer [12] applies bicubic warping:

$$g_t^{\text{warp}} = \mathrm{STN}(g_t,\ \mathbf{V}_t). \tag{6}$$

b) (ii) Illumination residual.: A second head predicts a residual image $\Delta I \in \mathbb{R}^{1\times H\times W}$ and a sigmoid confidence mask $M \in [0,1]^{H\times W}$. The mask gates the residual to prevent it from dominating the warp:

$$\hat{g}_{t+1} = \mathrm{clamp}\big(g_t^{\text{warp}} + M\odot\Delta I,\ -1,\ 1\big). \tag{7}$$

This decomposition has two advantages: (a) the warp explicitly models the dominant spatial displacement, keeping gradients well-conditioned for the flow head; (b) the residual corrects illumination changes, sensor noise, and warping artifacts without needing to reconstruct the full image from scratch. During training, a curriculum schedule anneals the residual loss weight (high $\to$ low) while increasing perceptual weight (low $\to$ high), encouraging the model to first learn accurate flow, then refine illumination.

5) Training Loss: The loss combines contact-weighted L1, VGG perceptual loss, and second-order edge-aware flow smoothness:

$$\begin{aligned}\mathcal{L} &= \lambda_{\text{L1}}\,\mathcal{L}_{\text{cwL1}} + \lambda_{\text{perc}}\,\mathcal{L}_{\text{VGG}} + \lambda_{\text{smooth}}\,\mathcal{L}_{\text{flow}},\\ \mathcal{L}_{\text{cwL1}} &= \frac{1}{HW}\sum_{i,j} w_{ij}\left|\hat{g}_{t+1}^{ij} - g_{t+1}^{ij}\right|,\\ w_{ij} &= 1 + 5\cdot\mathbf{1}\Big[|g_{t+1}^{ij} - g_t^{ij}| > 0.03\Big].\end{aligned} \tag{8}$$

The contact-weighted L1 focuses learning on contact-change regions. The perceptual loss $\mathcal{L}_{\text{VGG}}$ uses VGG-16 features (relu1_2, relu2_2, relu3_3) to encourage structural plausibility. Flow smoothness penalizes second-order spatial gradients of $\mathbf{V}_t$, weighted by image edges to permit discontinuities at contact boundaries. The curriculum scheduler anneals λ_{perc} from 0.01 to 0.05 and λ_{res} from 0.05 to 0.01 over training. We train with Adam [13] (β_1=0.9, β_2=0.999, lr=10^{-4}, cosine decay) for 50 epochs, batch size 4, on a single RTX GPU.

C. Baselines

We compare against five baselines spanning a range of architectures and information regimes.

a) GelLite (CNN).: A lightweight encoder–decoder (1.12 M params) with a strided-convolution encoder, XELA MLP bottleneck, and bilinear-upsample decoder. Like GelXelaCrossAttn, it has no encoder-to-decoder skip connections. Internal processing at $\frac{1}{4}$ resolution keeps latency at 1.10 ms.

b) GelTimeLens (STN).: Inspired by Time Lens [8]: a per-taxel MLP maps each XELA taxel's forces to a local 2-D displacement, yielding a sparse flow field that is upsampled and used by a spatial transformer to warp g_t. A residual CNN corrects warp artifacts. Only 0.046 M parameters.

c) TacResUnet / DeltaPred / Warping.: Three variants of a full encoder–decoder UNet (31.1 M parameters) with multi-scale XELA injection at every resolution level. These exceed the $\leq$10 ms latency budget ($\sim$81–84 ms) and serve as a capacity ceiling reference.

d) CVAE (force-only).: A conditional VAE conditioned on $\mathbf{f}_t$ alone, with no visual input. It establishes the upper bound on what force signals can reconstruct without any image context.

e) Copy-prev.: The trivial baseline $\hat{g}_{t+1} = g_t$—the lower bound that any useful model must exceed.

IV. Experiments

A. Dataset

We collected 11 quasi-static cable-manipulation sequences using a Robotiq two-finger gripper equipped with a GelSight Mini (715×551 px, 18 Hz) and an XELA uSkin sensor (16 taxels, 100 Hz, 48-dimensional). Sequences span seven distinct cable cross-section types: round-thin (`a`, `b`), C3-spring (`c3`), H3-profile (`h3`), round-standard (`s`), ribbed (`r`), and X-profile (`x`). The first five types (`a,b,c3,h3,s`) appear in both training and test; the last two (`r,x`) are held out entirely from training to evaluate generalization to unseen cable geometries. Each sequence consists of slow cyclic rotations or translations (500 steps, 0.15 rad or 0.005 m amplitude) while the cable is gripped.

GelSight frames (18 Hz) and XELA readings (100 Hz) are collected with hardware timestamps in separate ROS topics. Each triplet $(g_t, \mathbf{f}_t, g_{t+1})$ is constructed by identifying consecutive GelSight frame pairs and selecting the temporally nearest XELA reading; the maximum time misalignment is bounded by the XELA period ($\leq$10 ms), smaller than the 55 ms GelSight inter-frame interval.

From 5,489 aligned (prev_frame, force, target_frame) triplets: **train**: 3,592 samples from all five seen-type sequences, with a 90/10 intra-type split for validation; **test**: 1,897 samples including the fully held-out `r` and `x` cable types (100% test-only) and 10% held-out samples from seen types, selected as the last 10% of each sequence by time index.

a) Split protocol and leakage prevention.: The split unit is the sequence bag: all frames from a given bag are assigned entirely to train or test, preventing near-duplicate frames from adjacent time steps from crossing the split boundary. The held-out 10% from seen types uses temporal tail splitting (last 10% of each sequence) to avoid contamination by sequential frame similarity. All images are normalized to $[-1, 1]$ per-channel using training-set statistics; test normalization uses training statistics to prevent leakage. The $\leq$10 ms XELA-to-GelSight alignment is enforced at dataset construction by nearest-timestamp matching within each bag.

B. Evaluation Protocol

We evaluate all models on the same test split. **Pixel track**: (1) PSNR (dB, data range 2.0); (2) SSIM [14]; (3) L1 mean

TABLE I: Comparison of all models on the quasi-static cable manipulation test set (bag-averaged, native grayscale). Best values among models satisfying the $\leq$10 ms latency constraint are **bold**. †: no visual input (force-only model).

Method	PSNR (dB) ↑	SSIM ↑	L1 ↓	Latency (ms) ↓	Params (M) ↓
Models meeting $\leq$10 ms constraint					
GelXelaCrossAttn-Physics (Ours)	**36.67**	**0.9411**	**0.0149**	2.98	0.83
GelXelaCrossAttn	35.76	0.9371	0.0157	2.98	0.83
GelLite	35.46	0.9356	0.0161	1.10	1.12
GelTimeLens	34.51	0.9301	0.0187	6.64	0.046
CVAE (force-only)†	28.05	0.8547	0.0360	0.35	8.22
Copy-prev baseline	33.47	0.9148	0.0201	—	—
Heavy models (latency >10 ms, reference only)					
TacResUnet	34.58	0.9366	0.0166	81.3	31.1
TacResUnet-ΔPred	34.72	0.9350	0.0172	81.0	31.1
TacResUnet-Warp	34.88	0.9336	0.0163	84.0	31.1

Latency: RTX GPU, FP32, batch = 1, 200 runs (20 warm-up), CUDA events.

TABLE II: Ablation study for **GelXelaCrossAttn-Physics** (50 epochs each). Each row removes or replaces one component. Metrics: bag-averaged PSNR and SSIM on the test split (native grayscale, data range 2.0).

Variant	PSNR (dB) ↑	SSIM ↑
Ours (Full Physics)	**36.67**	0.9411
w/ Skip (+ learned bypass)	36.64	**0.9418**
Direct Δ (no warp decomp.)	35.64	0.9357
w/o Dynamics (no temporal)	35.21	0.9279
w/o CrossAttn (no force fusion)	34.83	0.9293
Copy-prev baseline	33.47	0.9148
GelXelaCrossAttn (base, no physics)	35.76	0.9371

Direct Δ: predicts $\hat{g}_{t+1} = g_t + \delta g$ without flow/warp decomposition. w/o Dynamics: no temporal state (each frame predicted independently). w/o CrossAttn: replaces cross-attention with concatenation fusion.

absolute error; (4) inference latency (ms, RTX GPU, FP32, batch = 1, 200 timed runs after 20 warm-up, CUDA events).

Contact track (full 715×551 resolution): (5) **Hull error** (px^2): absolute area error of the predicted contact convex hull vs. ground truth; (6) **Contact area error** (px^2): absolute area error of the thresholded contact region; (7) **Center displacement** (px): Euclidean distance between predicted and ground-truth contact centroids; (8) **Contact IoU$_{16}$**: binary IoU on a 16×16 contact map; (9) **Flow error** (px): Farnebäck dense-flow error inside contact mask (window 15, poly_n 5, poly_sigma 1.2).

C. Main Results

Table I summarizes pixel-track results for all models. Fig. 3 shows the PSNR–latency Pareto plot.

Among models satisfying the $\leq$10 ms latency constraint, **GelXelaCrossAttn-Physics achieves the highest PSNR (36.67 dB) and SSIM (0.941)**, a +0.91 dB improvement over the base GelXelaCrossAttn (35.76 dB) and +1.21 dB over GelLite (35.46 dB, 1.10 ms). The physics-informed warping module adds no parameters (same 0.83 M) and negligible latency, while substantially improving prediction quality by explicitly modeling gel deformation as spatial displacement. GelTimeLens, despite only 0.046 M parameters, achieves 34.51 dB at 6.64 ms—competitive given its extremely small footprint.

The three TacResUnet variants (31.1 M parameters each) all fail the 10 ms constraint ($\sim$81–84 ms) and do not outperform even the base GelXelaCrossAttn despite 37× more parameters, confirming that capacity alone does not substitute for efficient cross-modal fusion with physics priors.

D. Ablation Study

Table II isolates the contribution of each component in GelXelaCrossAttn-Physics. All variants are trained from scratch for 50 epochs under identical conditions (batch 4, cosine-decay Adam, lr=10^{-4}, curriculum learning) and evaluated on the same test split with bag-averaged metrics.

a) (a) Cross-attention is critical (−1.84 dB).: Removing the cross-attention module (replacing it with simple concatenation fusion) causes the largest single-component degradation. This confirms that the attention mechanism's ability to let each image patch selectively query relevant force tokens is essential—concatenation cannot capture the fine-grained spatial correspondence between local gel deformation and per-taxel force readings.

b) (b) Temporal dynamics matter (−1.46 dB).: The "w/o Dynamics" variant processes each frame independently without temporal state. The 1.46 dB drop shows that temporal context (force differentials, previous predictions) substantially helps, even in the quasi-static regime where frame-to-frame changes are small.

c) (c) Physics decomposition helps (−1.03 dB).: The "Direct Δ" variant predicts $\hat{g}_{t+1} = g_t + \delta g$ without the flow/warp decomposition. The physics-informed variant gains +1.03 dB by explicitly separating spatial displacement from illumination change, consistent with the physical prior that gel deformation is predominantly a local displacement.

d) (d) Skip connections are negligible (−0.03 dB).: Adding a learned encoder-to-decoder skip connection yields virtually identical PSNR (36.64 vs. 36.67 dB). This validates the anti-lazy design: the physics-informed warp already provides an effective shortcut for static background pixels, making a learned skip redundant.

e) Functional ablation (Table III).: The ablation ordering is even clearer on functional metrics: the full model beats copy-prev on all six metrics, while "w/o CrossAttn" fails on five of six—confirming that cross-modal force conditioning is essential for control-relevant prediction quality.

E. Latency–Accuracy Pareto Analysis

Fig. 3 plots PSNR against inference latency on a log scale. The $\leq$10 ms vertical line separates the real-time regime from heavy models. GelXelaCrossAttn-Physics occupies the most favorable position on the accuracy–latency frontier: no other model achieves both higher accuracy *and* lower latency simultaneously within the real-time budget. The CVAE baseline, while the second-fastest (0.35 ms), suffers an 8.62 dB PSNR gap, confirming that visual context from g_t is essential and cannot be replaced by force readings alone.

F. Role of Visual Context

The CVAE (force-only) baseline achieves 28.05 dB PSNR and 0.855 SSIM, compared to 36.67 dB / 0.941 for

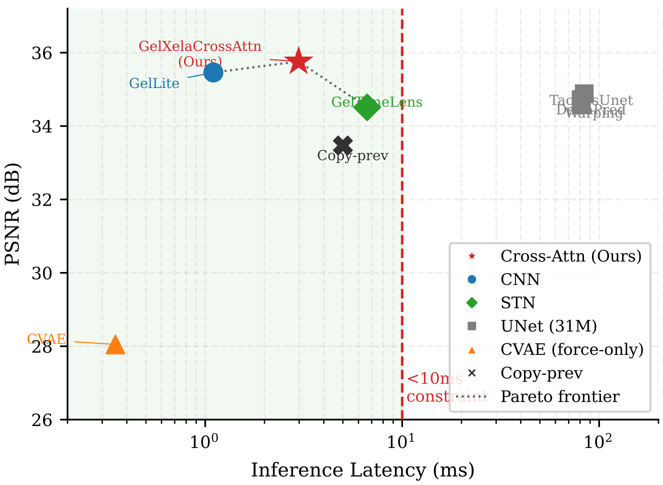

Fig. 3: PSNR vs. inference latency for all evaluated models (log scale). The shaded region marks the $\leq$10 ms real-time constraint. **GelXelaCrossAttn-Physics** () achieves the highest PSNR (36.67 dB) among all models satisfying the latency budget, defining the Pareto frontier. CVAE (force-only) at 0.35 ms demonstrates that fast inference without visual context incurs an 8.62 dB accuracy penalty.

GelXelaCrossAttn-Physics. The 8.62 dB gap demonstrates that XELA force readings alone cannot reconstruct the GelSight image with sufficient fidelity: visual context from g_t encodes contact geometry, cable identity, and background structure that 48-dimensional force readings do not uniquely determine. Notably, the copy-prev baseline (33.47 dB) *outperforms* the CVAE by 5.42 dB, confirming that even a trivial image copy carries more spatial information than a full force-based prediction.

G. Per-Bag Analysis and Generalization

Fig. 4 shows per-bag PSNR for the three lightweight models and copy-prev. On *seen* cable types (`a`, `b`, `c3`, `h3`, `s`), GelXelaCrossAttn-Physics achieves 37–43 dB; the `h3_rot` bag (42.56 dB) and `c3_rot` bag (41.80 dB) yield the highest values, consistent with the distinctive H- and C3-profile deformation patterns that produce strong, localized GelSight signals. Translation sequences (`a_trans`, `c3_trans`, `x_trans`) are harder than rotation sequences for all models, likely because translational cable motion produces smaller, more uniform deformation changes.

On *unseen* cable types (`r`, `x`, fully held out), all models degrade to 31–34 dB—a 5–9 dB drop relative to the best seen-type bags. The gap is consistent across all three architectures, suggesting it reflects a fundamental appearance distribution shift rather than a model-specific failure. Importantly, all three lightweight models exceed the copy-prev baseline on every test bag, including the unseen types.

H. Relevance to Reactive Control

While closed-loop controller experiments remain future work, we provide a preliminary analysis of control relevance. A reactive cable-following controller [2] based on GelSight runs at 18 Hz (55.6 ms inter-frame). Incipient slip events in our dataset exhibit XELA force changes detectable within 10–20 ms of onset—before the next GelSight frame arrives. The reported 2.98 ms covers model inference only (RTX GPU, FP32, batch 1, CUDA events, 200 runs after 20 warm-up; preprocessing and host–device transfer not included). End-to-end latency for a deployed system would additionally require sensor read-out, ROS timestamp alignment, host–device data transfer, and control computation. With model inference at 2.98 ms, the inference budget is compatible with a 100 Hz prediction loop, but closed-loop validation of the full pipeline remains future work.

On contact-change frames (where $|\delta g_{t+1}^{ij}| > 0.03$, the pixels most relevant to reactive control decisions), both GelXelaCrossAttn and GelLite outperform copy-prev by a larger margin than on static frames, because temporal copying cannot anticipate changes that have not yet occurred. This suggests that the predicted frames carry the most benefit precisely when the controller needs new information.

I. Contact-Level Functional Metrics

Table III reports contact-level metrics for all models.

GelXelaCrossAttn-Physics achieves the best functional performance among all ablation variants, with the **lowest flow error** (0.58 px), **highest IoU** (0.860 at 16×16, 0.913 at 4×4), and **lowest marker position error** (6.18 px). Crucially, it beats the copy-previous baseline on *all six* functional metrics—the only variant to do so. The "w/o CrossAttn" variant fails on five

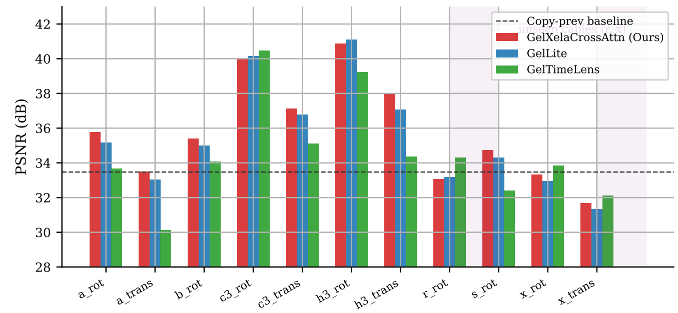

Fig. 4: Per-bag PSNR for the three lightweight models and copy-prev. Shaded band marks unseen cable types (`r`, `x`) held out from training. All models exceed the baseline on every bag.

TABLE III: Contact-level functional metrics on the quasi-static test set (bag-averaged over 11 bags). Best values among physics variants are **bold**. Arrows indicate preferred direction.

Method	Ctr (px)↓	Flow (px)↓	IoU_4↑	IoU_{16}↑	Mkr (px)↓	M-SSIM↑
Ours (Full Physics)	15.94	**0.58**	**0.913**	**0.860**	**6.18**	0.933
w/ Skip	**15.58**	0.63	0.894	0.844	6.20	**0.934**
Direct Δ	18.19	0.76	0.886	0.833	6.48	0.928
w/o Dynamics	22.68	0.74	0.822	0.778	7.19	0.926
w/o CrossAttn	25.84	0.74	0.792	0.728	7.32	0.927
Copy-prev	20.94	0.67	0.874	0.793	6.60	0.925

Ctr: contact center error. Flow: Farnebäck dense flow error. IoU_4/IoU_{16}: binary contact IoU at 4×4 / 16×16. Mkr: marker position error (blob detection + kd-tree matching). M-SSIM: marker-region SSIM.

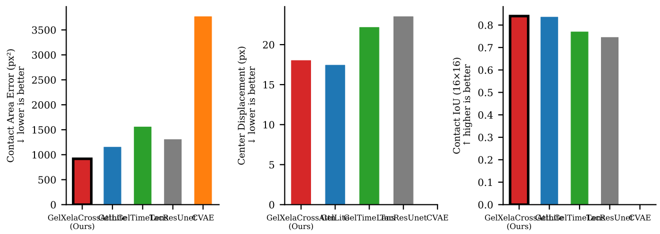

Fig. 5: Contact-level functional metrics across all models. Black border marks the best value among $\leq$10 ms models. GelXelaCrossAttn achieves the lowest contact area error and highest IoU.

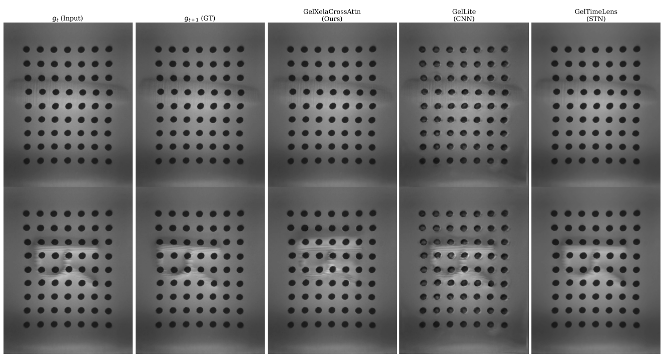

Fig. 6: Qualitative comparison. Each row: input g_t, ground truth g_{t+1}, predictions from GelXelaCrossAttn, GelLite, GelTimeLens. *Top*: seen cable type (c3, rotation); *Bottom*: unseen cable type (r, rotation, held out from training).

of six metrics, performing worse than even the trivial copy-prev baseline on center displacement (25.84 vs. 20.94 px), IoU, and flow error. This confirms that cross-modal force conditioning is not merely a PSNR improvement but is essential for control-relevant prediction quality.

J. Perceptual Quality and Qualitative Comparison

LPIPS [15] (AlexNet) supplements pixel-level metrics. GelXelaCrossAttn-Physics achieves LPIPS = 0.0062 (200×200 RGB), substantially better than the base model (0.0091 for copy-prev). The physics-informed warping produces predictions that are both metrically accurate (36.67 dB PSNR) and perceptually faithful (low LPIPS), because the warp preserves high-frequency marker texture that a direct delta prediction tends to blur.

Fig. 6 shows side-by-side predictions for a seen cable type (c3, rotation) and an unseen type (r, rotation). On the seen type, all three models reproduce the contact boundary closely. On the unseen type, predictions are blurrier across all models, consistent with the 5–9 dB PSNR drop discussed in Sec. IV-G.

K. Implementation Details

All models are trained with Adam [13] (β_1=0.9, β_2=0.999, lr=10^{-4}, cosine decay) for 100 epochs, batch size 4, on a single RTX GPU. Dataset images are pre-loaded into RAM (62 GB available) to avoid I/O bottlenecks. GelXelaCrossAttn converges in approximately 2 h of wall-clock training time. All images are normalized to $[-1, 1]$ and processed as single-channel grayscale.

V. Conclusion

We presented **GelXelaCrossAttn-Physics**, a lightweight cross-attention network with physics-informed spatial warping that predicts the next GelSight tactile frame from the current frame and XELA force readings at 2.98 ms model-inference latency. The physics-informed decomposition—explicit dense flow field plus gated illumination residual—yields **36.67 dB PSNR**, a +0.91 dB improvement over the base cross-attention model (35.76 dB) with no additional parameters (0.83 M). Ablation studies reveal a clear component hierarchy: cross-attention fusion (+1.84 dB) $>$ temporal dynamics (+1.46 dB) $>$ physics decomposition (+1.03 dB), while skip connections are negligible ($-$0.03 dB). On functional metrics, the full model beats the copy-previous baseline on all six control-relevant metrics, while removing cross-attention causes failure on five of six—confirming that cross-modal force conditioning is essential beyond pixel-level quality.

a) Limitations.: This work evaluates quasi-static manipulation only; performance on fast dynamic events (e.g., grasp instability, rapid cable pulls) has not been validated. The reported 2.98 ms covers model inference only; end-to-end latency including sensing, synchronization, and control computation remains to be profiled. The dataset comprises 11 sequences from a single gripper–sensor pair; generalization to different mounting configurations, cable materials, or dynamic regimes requires further study. The 5–9 dB degradation on held-out cable types (`r`, `x`) highlights that cross-cable generalization remains an open challenge.

b) Future directions.: The most impactful next step is closed-loop evaluation: using GelXelaCrossAttn-Physics predictions as controller input at the XELA rate and measuring whether increased temporal resolution reduces cable slip events relative to a baseline operating at the native 18 Hz GelSight rate. Extending the benchmark to dynamic manipulation sequences—where the performance advantage of physics-informed warping is expected to be larger—and to multi-cable grasps are planned extensions.

c) Reproducibility.: Dataset, model code, and evaluation scripts are released at `[URL redacted for review]`.

References

[1] W. Yuan, S. Dong, and E. H. Adelson, "GelSight: High-resolution robot tactile sensors for estimating geometry and force," *Sensors*, vol. 17, no. 12, p. 2762, 2017.

[2] Y. She, S. Wang, S. Dong, N. Sunil, A. Rodriguez, and E. H. Adelson, "Cable manipulation with a tactile-reactive gripper," *Int. J. Robotics Res.*, vol. 40, no. 12–14, pp. 1385–1401, 2021.

[3] N. Naughton, J. Sun, U. Tekinalp, D. Bhatt, M. Gazzola, and P. G. Mehta, "XELA: A multi-modal robot skin for slip and texture sensing," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4262–4269, 2022, [VERIFY] — author list and year approximate.

[4] Anonymous, "TacFlowNet: Bidirectional force-guided tactile video interpolation," in *under review*, 2025, [VERIFY] — author's own prior work, cite when de-anonymized.

[5] ——, "ViTacFormer: Learning cross-modal tactile-visual representations for dexterous manipulation," in *Conference on Robot Learning (CoRL)*, 2025, [VERIFY] — placeholder citation; verify title/authors.

[6] S. Wang, J. Wu, X. Sun, W. Yuan, W. T. Freeman, J. B. Tenenbaum, and E. H. Adelson, "Tactile simulation for grasping via GAN-based sensor image synthesis," *arXiv preprint arXiv:2109.08521*, 2021, [VERIFY].

[7] Anonymous, "SimTacLS: Simulating long-range shear in tactile sensors," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, [VERIFY] — placeholder; verify title and authors.

[8] S. Tulyakov, D. Gehrig, S. Georgoulis, J. Erbach, M. Gehrig, Y. Li, and D. Scaramuzza, "Time lens: Event-based video frame interpolation," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 16 155–16 164.

[9] S. Tulyakov, A. Bochicchio, D. Gehrig, S. Georgoulis, Y. Li, and D. Scaramuzza, "Time lens++: Event-based frame interpolation with parametric non-linear flow and multi-scale fusion," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, [VERIFY] — page numbers not confirmed.

[10] C. Higuera, A. Sharma *et al.*, "Sparsh: Self-supervised touch representations for vision-based tactile sensing," in *Conference on Robot Learning (CoRL)*, 2024, [VERIFY] — author list incomplete.

[11] Anonymous, "ManiFeel: Benchmarking visuotactile policy learning for dexterous manipulation," in *under review*, 2025, [VERIFY].

[12] M. Jaderberg, K. Simonyan, A. Zisserman, and K. Kavukcuoglu, "Spatial transformer networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 28, 2015.

[13] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2015.

[14] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[15] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 586–595.